

Approach to children and adolescents with suspected eating disorder/disordered eating behaviors

- **Suspect eating disorder when patient present with the following symptoms:**
 - Precipitous weight changes
 - Sudden changes in eating behaviors (new vegetarianism/veganism, gluten-free, lactose-free, elimination of certain foods or food groups, eating only “healthy” foods, uncontrolled binge eating)
 - Sudden changes in exercise patterns, excessive or compulsive exercise or involvement in extreme physical training
 - Body image disturbance, the desire to lose weight despite low or normative weight, or extreme dieting behavior regardless of weight
 - Abdominal complaints in the context of weight loss behaviors
 - Electrolyte abnormalities without an identified medical cause (especially hypokalemia, hypochloremia, or elevated bicarbonate)
 - Hypoglycemia
 - Bradycardia
 - Syncope
 - Amenorrhea or menstrual irregularities
 - Type 1 diabetes mellitus with poor glucose control or recurrent diabetic ketoacidosis (DKA) with or without weight loss
 - Avoidant, restrictive eating resulting in weight loss or poor weight gain
- **Medical instability** can occur across a range of weights in adolescents with eating disorders with **rapid weight loss** indicator of medical risk regardless of BMI
- Medical stabilization, nutritional rehabilitation for early weight restoration to address nutrient deficiencies, management of refeeding and its potential complications, and interruption of purging/compensatory behaviors are the **immediate goals of treatment** for all patients with Eating disorder or disordered eating behaviors regardless of their weight or BMI
- For patients presenting with weight loss, consider following (if other weight/BMI data available)
 - Calculate percent median BMI = current BMI/BMI at 50th percentile
 - Calculate percent body weight loss in last 12 months or last 6 months
 - Categorize degree of malnutrition

Term	Abbreviation	Data required and/or Calculation		
Body Mass Index	BMI	Weight (kg) / [height (m)] ²		
Median BMI	mBMI	50 Percentile BMI for age and sex		
Percent m BMI	%m BMI	(Current BMI / mBMI) x 100		
	Mild	Moderate	Severe	
%m BMI	80 – 90%	70 – 79%	<70%	
BMI z score	-1 to -1.9	-2 to -2.9	-3 or Greater	
Magnitude of Weight loss	5%	7.5%	10%	
Rapidity of weight loss		5% in 1 month 7.5% in 3 months 10% in 6 months 20% in 1 year	>5% in 1 month > 7.5% in 3 months >10% in 6 months >20% in year	

Golden, J Adolescent Health 2022

- Obtain Labs- CBC, CMP, Magnesium, phosphorus, TSH With patients presenting with abdominal symptoms or purging behaviors or vomiting consider obtaining amylase and lipase
- Obtain EKG and orthostatic vitals
 - CDC- <https://www.cdc.gov/steady/media/pdfs/STEADI-Assessment-MeasuringBP-508.pdf>
- Assess for refeeding risk

At Risk	Children & adolescents: % m BMI- 80-90 Very little intake for greater than 5 days Adult <18.5 BMI
High Risk	One or more of the following: • Adult - BMI<16 kg/m² • Children and Adolescents: % m BMI 70-79 • Unintentional weight loss ≥15% within the previous 3 - 6 months • Very little nutritional intake > 10 days • Low levels of potassium, phosphate, or magnesium prior to feeding Two or more of the following: • Adult: BMI <18.5 • Unintentional weight loss ≥10% within the previous 3 - 6 months • Very little nutritional intake for > 5 days • Hx of alcohol abuse, on drugs including insulin, chemotherapy, antacids, diuretics for weight loss
Extremely High risk	Adults: BMI ≤14 kg/m ² Children & adolescents: % m BMI less than 70 Negligible intake for greater than 15 days

Ref- Academy of eating disorder -Guidebook for nutrition treatment of eating disorder

- Serum electrolytes (Sodium, potassium, phosphorus, magnesium) may be normal prior to initiating nutrition

- Treatment of eating disorder occurs in variety of clinical settings based on severity of symptoms, degree of malnutrition, medical stability, co-morbid mental health conditions, presence or absence of recovery and supportive environment and access to care.
- **Goal of inpatient medical management** is initiating nutrition in a structured setting with close monitoring for refeeding, interrupting eating disorder behaviors, educating parents/guardian about diagnosis and treatment and facilitate transition to appropriate levels of care.
- Inpatient nutritional rehabilitation and medical stabilization is indicated for patient presenting with 1 or more criteria (See Detailed Admission Criteria)

Admission Criteria for inpatient medical management

Meets at least 1 or more criteria from critical/red

Meets at least 2 or more from Severe/Amber

For any questions related to admission, consult with Peds Hospitalist team. Please consider collaborating with Adolescent Medicine provider

System	Measure	CRITICAL (RED)	SEVERE (AMBER)
Cardiovascular	HR	<40 awake, < 35 asleep	<50 awake, < 40 asleep
	BP	<90/45 awake	<90/60 awake
	Orthostatic changes	HR increase ≥ 40 ; SBP drops by 20 mm Hg or more; DBP drops by 10 mm Hg or more	SBP drop between 10-20 mm or DBP drops ≤ 10 mm Hg with symptoms; HR increase by ≥ 30 beats/min without changes in BP but with observable symptoms such as dizziness
EKG	Arrhythmias	Any arrhythmia other than sinus tach, sinus brady or sinus arrhythmia	
	QTc	> 0.475	0.440 to 0.475
Electrolytes	Phosphorus	< 3.0	3.0 to 4.0 (below threshold for age)
	Potassium	< 2.5	2.5 to 3.0
	Sodium	<130	130 to 135
	Magnesium	≤ 1.5	1.6-1.8
Dehydration		Severe (> 10%) (reduced urine output, dry mouth, postural BP drop (see above), decreased skin turgor, sunken eyes, tachypnea, tachycardia	Moderate (5 to 10%) reduced urine output, dry mouth, postural BP drop (see above), normal skin turgor, some tachypnea, some tachycardia, peripheral edema

Other organs		End-organ damage, Mallory Weiss tears, myocarditis, pancreatitis	Syncope, transaminitis,
Other psychiatric conditions- Consult with Child Psychiatry on call		SI/HI associated with eating disorder	Moderate-Severe OCD associated with eating disorder
			Moderate-Severe depression associated with eating disorder
Body mass index (BMI)	% median for age and sex	< 70 % median BMI	70 to 80 % median BMI
Percent of weight loss (Degree of malnutrition)	Weight loss/Highest weight	Recent loss of weight of $\geq 1\text{kg/week}$ for 2 consecutive weeks or $\geq 20\%$ body mass loss in 12 months or $\geq 10\%$ Body mass loss in 6 months	15%-20% body weight loss in 12 months 7.5%-10% in 6 months
Food refusal	Total food refusal	Acute food refusal for 2+ days	Acute x 1 to 2 days
	Usual daily calorie intake	400 to 600	600 to 1000
Purging/vomiting/laxative abuse		Multiple daily episodes of vomiting daily and frequent use of laxatives.	Multiple episodes >3 times per week or laxative use multiple times a week
Behavior around food		Violent	Active resistance
Exercise		High levels of dysfunctional exercise in the context of malnutrition (>2h/day)	2 to 3 hours/day Moderate levels of dysfunctional exercise in the context of malnutrition (>1h/day)
Engagement with care plan		Physical struggles with parents/care-givers over nutrition or reduction of exercise Harm to self Poor insight or motivation Parents/care- givers unable to implement meal plan prescribed Failure of intensive outpatient treatment	Poor insight or motivation Resistance to weight gain. Parents/care-giver unable to implement meal plan prescribed Failure of outpatient treatment- fail to make progress in 6-8 weeks i.e not gaining 0.5-1lbs per week or continuing to lose weight or unable to reduce purging, bingeing, laxative use by 50% in 8-10 sessions resulting in medical complications

References: -

- [Identification and Management of Eating Disorders in Children and Adolescents](#)
- [Medical Emergencies in eating disorder- NICE guidelines](#)
- [Medical Management of Restrictive Eating Disorders in Adolescents and Young Adults](#)
- [Medical instability in typical and atypical adolescent anorexia nervosa: systematic review and meta-analysis](#)
- [The American Psychiatric Association Practice Guideline for the Treatment of Patients with Eating Disorders](#)

Diagnostic criteria for eating disorders (DSM V)

Anorexia Nervosa	Bulimia Nervosa	Binge eating	Other specified eating disorder	ARFID
• Restricted caloric intake relative to energy requirements, leading to significantly low body weight for age, sex, projected growth, and physical health • Intense fear of gaining weight or behaviors that consistently interfere with weight gain, despite being at a significantly low weight • Altered perception of one's body weight or shape, excessive influence of body weight or shape on self-value, or persistent lack of acknowledgment of the seriousness of one's low body weight	Repeated episodes of binge eating followed by compensatory behaviors such as self-induced vomiting/purging, laxative use, diuretics, or other medication or excessive exercise On average, the binge eating and compensatory behaviors both occur at least once a week for 3 months. Self-value is overly influenced by body shape and weight	Binge eating is characterized by both of the following: within a distinct period of time (eg, 2 h), eating an amount of food that is clearly larger than what most individuals would eat during a similar period of time under similar circumstances and a sense that one cannot limit or control their overeating during the episode. The binge-eating episodes include 3 or more of the following: eating much more quickly than normal, eating until uncomfortably full, eating large amounts of food when not feeling hungry, eating alone because of embarrassment at how much one is eating, and feeling guilty, disgusted, or depressed afterward No compensatory behaviors	Atypical AN: all of the criteria for AN are met yet the individual's weight is within or above the normal range despite significant weight loss BN (of low frequency and/or limited duration): All of the criteria for BN are met, but, on average, the binge eating and compensatory behaviors occur less than once a week and/or for <3 months BED (of low frequency and/or limited duration): All of the criteria for BED are met, but, on average, the binge eating occurs less than once a week and/or for <3 mo Purging disorder: recurrent purging behavior (eg, self-induced vomiting; misuse of laxatives, diuretics, or other medications) in the absence of binge eating with the intent to influence weight or body shape	A disrupted eating pattern (eg, seeming lack of interest in eating or food; avoidance based on the sensory qualities of food; concern about unpleasant consequences of eating) as evidenced by persistent failure to meet appropriate nutritional and/or energy needs associated with 1 (or more) of the following: significant weight loss or, in children, failure to achieve expected growth and/or weight gain, marked nutritional deficiency, reliance on enteral feeding or oral nutritional supplements, significant interference with psychosocial functioning The disturbance cannot be better explained by lack of available food or by an associated culturally sanctioned practice or attributed to a coexisting medical condition nor better explained by another mental disorder.